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Isolated bidirectional DC-DC Converter: A topological review

Mohammad Aslam Alam [a,*], Ahmad Faiz Minai [a], Farhad Ilahi Bakhsh [b]

[a] Department of Electrical Engineering, Integral University Lucknow, India
[b] Department of Electrical Engineering, National Institute of Technology Srinagar, J&K, India

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ABSTRACT

Bidirectional DC-DC converters (BDCs) are certainly an important power electronic converter for managing bidirectional power flow in various applications. It offers the ability to flow power in both directions, which is useful in systems with renewable energy sources and energy storage. BDCs are becoming increasingly important in various applications, including renewable energy systems, electric vehicles, and energy storage. The development of new BDC topologies and control strategies is expected to further improve their efficiency and performance. Non-Isolated Bidirectional DC-DC Converters (NBDCs) and Isolated Bidirectional DC-DC Converters (IBDCs) are the two main types of BDCs with different isolation properties and trade-offs in terms of size, cost and performance. Topological classification of BDCs further distinguishes various BDC types based on their configurations and voltage-boosting techniques. IBDCs are preferred over NBDCs due to high-power applications, and wide voltage range with higher efficiency; the performance of IBDCs is improving day by day. To provide a framework for comparison in implementing novel configurations or identifying the best converter for a given application, this study is concentrated on each topology and its usual applications. Soft switching converters are also discussed pointed to Resonant Power Converters (RPCs) for the application into Electric Vehicles (EVs). It is worth noting that the choice of BDC depends on the specific requirements of the application, such as power level, voltage range, efficiency, and cost. A detailed comparison, application, advantage and disadvantage table can be useful in comparing the characteristics of different BDCs and selecting the most suitable one.

1. Introduction

Recent development in power systems using renewable energy such as Hybrid Vehicles, renewable energy-based systems bought various challenges. Converters are interfaced in between the distributed generator and dc bus but demand is continuously increasing; so to fulfil the load demand researchers focused on (a) Increasing voltage level (b) efficiency and (c) size reduction of the converter [1]. Traditionally unidirectional converters were used for the power transfer from input to output and vice versa but during the integration of renewable energy sources and energy storage systems; number of electric components increases consequently its noise, size and heat consumption also increases. Also, during input voltage variation, voltage and current stress introduced. To overcome these challenges bidirectional converters are used, in which unidirectional switches (single-quadrant switches) are replaced with bidirectional switches (two-quadrant switches) [2] that introduced Bidirectional Converter (BDC). By implementing a Bidirectional Converter (BDC) it acts in both modes i.e., buck and boost as per requirement of system. It is also used for interfacing Energy Storage System (ESS) with Renewable Energy System (RES) and in various industries like transport, aerospace applications, system performance and efficiency are increased whereas the size of the system decreased because only one converter is used instead of two individual's converters. It is used in various hybrid systems such as fuel cells and solar PV for supplying the demand of both DC as well as AC load by using an inverter into consideration. These BDCs are categorized into Non-Isolated Bidirectional DC-DC Converters (NBDCs) and Isolated Bidirectional DC-DC Converters (IBDCs) and further these are grouped according to their configurations and voltage-boosting methods. Basic configuration of NBDCs are buck-boost, Cuk, SEPIC and ZETA converters offer simple and cost-effective solutions for low-power applications. Moreover, advanced configurations are switched capacitor, cascaded, interleaved and multilevel bridges are suitable for high-power applications and offer higher efficiency, lower switching losses, and easy control methods. For IBDCs basic configurations are Flyback, Cuk, Push-pull and Forward converter. It provides easy-to-use, cost-effective solutions

* Corresponding author.
E-mail address: aslamintegral@gmail.com (M.A. Alam).

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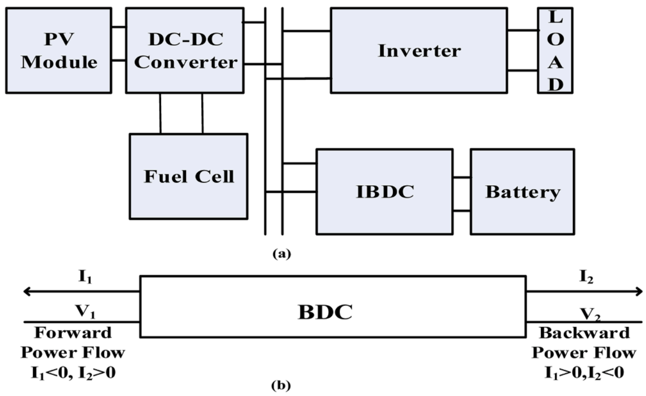

Fig. 1. **(a)** Load Demand Management using IBDC **(b)** Power flow in BDC.

for moderate voltage applications. Additionally, advanced configurations are Half-Bridge, full-bridge, Dual Active Bridge and Multiport bridges are designed for high-power applications as they offer improved efficiency, reduced switching losses and simple control strategies. As shown in Fig. 1(a) by using BDCs we can store power into batteries or supercapacitors which is termed as Forward Mode (Forward Power Flow) and in Backward mode (Backward Power Flow) these stored power can be used during unavailability or overload conditions. In Fig. 1 (b) value of current I_1 is less than zero which shows the availability of a primary source. At this stage, a bidirectional converter allows to store in ESS. When primary source is unavailable i.e. I_1 is greater than zero bidirectional converters allow to flow from ESS to the load. This paper is organized as Section-1 Introduction, Section-2 Classification of BDC, Section–2.1 Non-Isolated Bidirectional Converter (NBDC), Section-2.2 Isolated Bidirectional dc-dc converter (IBDC), Section-3 Resonant Converters and Section-4 Conclusion.

2. Classification of bidirectional converter

Mainly Bidirectional DC-DC Converter (BDC) converters are subdivided as Non-Isolated & Isolated Bidirectional converters. NBDCs transmits power in absence of magnetic isolation which means it doesn't use a transformer for the power exchange which is advantageous in various applications over IBDC where size and weight are a major concern but it has the demerit of low voltage gain and simpler configuration whereas Isolated Bidirectional Converter is employed in various power applications due to its efficiency, reliability, lightweight and compactness. When DC is converted to AC and again rectified into DC after transferring through a high-frequency transformer but it suffers from leakage inductance and magnetic interference [3]. When efficiency and high voltage gain is not an important parameter, for these applications non-isolated bidirectional converters without magnetic coupling can be useful, it comprises only switching devices and passive components but for high voltage gain magnetic coupling is utilized which betters its efficiency and reliability. As discussed in previous section NBDCs are classified based on basic configuration and voltage boosting techniques as shown in Fig. 2. Basic configurations are derived from simpler structure of bidirectional converter whereas voltage boosting techniques are obtained by rearranging this simpler structure. Detailed

Table 2
Advantage and Disadvantages of NBDC.

Advantages	Disadvantages
1. Balanced structure	1. Operates only in buck mode or in boost mode
2. Low ripple current on both side	2. For higher voltage ratio its design(structure) is impractical.
3. Wide operating range of voltage	3. Less galvanic isolation between the two sides.
4. Simple driver circuitry	4. Higher voltage gain requires an excessive duty cycle ratio.
5. Safety from short circuit	5. Discontinuous output current

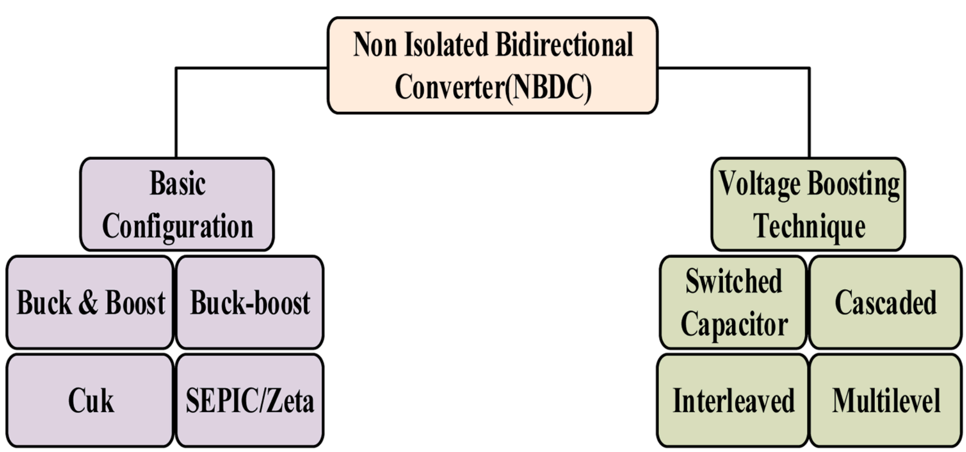

Fig. 2. General classification of Non-Isolated Bidirectional topologies.

Table 1
Comparison of NBDC topologies.

Reference		Topology	Inductors	Capacitors	Switches
[5]	Basic Configuration	Basic Buck-Boost	1	2	2
[6]		Buck-Boost	2	2	2
[7]		Cuk	2	3	4
[8]		SEPIC/Zeta	2	3	2
[9,10]	Advance Configuration	Switched Capacitor	1	2	4
[11]		Cascaded	0	3	4
[12]		Interleaved	$n = 2$	2	2n=4
[13]		Multilevel	0	$\frac{n(n+2)}{2}$ =6	$n(n+1)$ =12

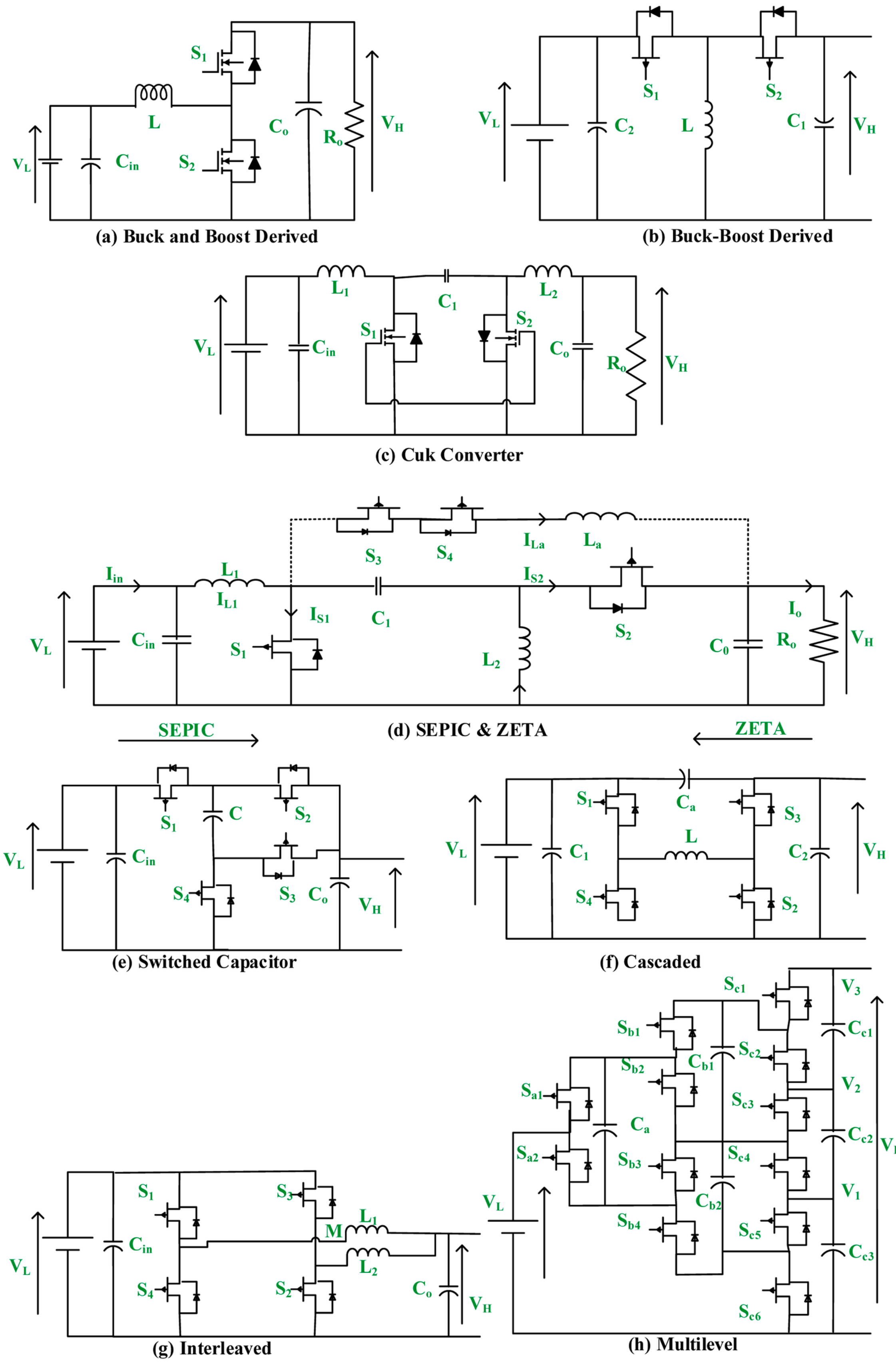

Fig. 3. Basic and bridge topology of Non-Isolated Bidirectional Converter.

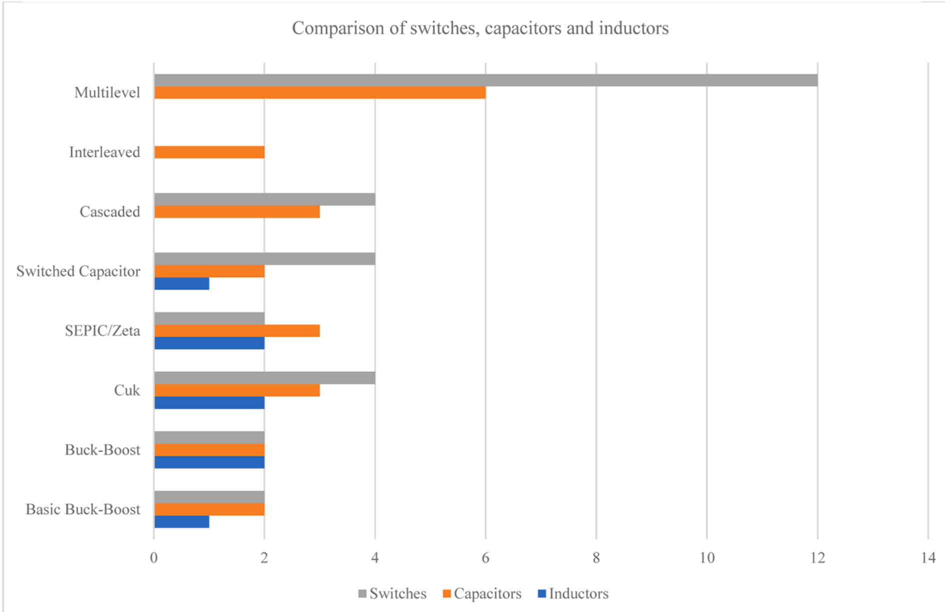

Fig. 4. Comparison of NBDCs in terms of switches, capacitors and inductors.

comparison of NBDCs has been done in [4] . Table 1 and Fig. 4 illustrates the application and comparison of all Non-Isolated Bidirectional Converters (NBDCs) which include the number of inductors, switches and capacitors. Also, advantage and disadvantage of NBDCs are discussed in Table 2.

2.1. Non-Isolated bidirectional converter

2.1.1. Basic configuration

2.1.1.1. Buck & boost converter. It is an arrangement of a step-down converter in anti-parallel with a step-up converter to lower and higher voltage respectively [5] but traditionally unidirectional switches like MOSFETs and diode were used for the operations, later these unidirectional switches are replaced with bidirectional switches, due to its unidirectional power flow and it also does not provide any current conduction path for the diode during reverse direction but for NBDCs it is obtained by replacing unidirectional switches with bidirectional switches as illustrated in Fig. 3(a). When output voltage (V_H) is higher as compared to input voltage (V_L); this operation is known as boost mode and vice versa is buck mode.

2.1.1.2. Buck-Boost converter. In Fig. 3(b) conventional Buck-boost converter is shown with bidirectional switches. For high-power application effect of a diode during step down minimizes efficiency of the circuit whereas we can use a diode in different applications. In Boost mode, output voltage (V_H) is higher as compared to input voltage (V_L) & vice-versa for buck mode but polarity of output voltage is altered which causes ripple. To neglect these issues more switches are added to circuit [6].

2.1.1.3. Cuk converter. Cuk converter is obtained by replacing main diode with MOSFET. Continuous input and output current is delivered by cuk converter. Coupled inductor is employed which eliminates current ripples in input/output of converter. So Cuk converters are interfaced with energy storage system [7] in Fig. 3(c) boost and buck configuration that are in series with energy storage capacitor which allows for both higher and lower output voltages [14].

2.1.1.4. SEPIC/Zeta converter. SEPIC and ZETA converter is obtained by rearranging Cuk converter as illustrated in Fig. 3(d). These converters both higher (V_H) and lower voltages (V_L) and polarity of DC bus is similar. When power flow from lower voltage (V_L) to higher voltage (V_H) converter act as SEPIC and for vice versa it acts as ZETA [8].

2.1.2. Voltage boosting technique

2.1.2.1. Switched capacitor. Switched Capacitor (SC) is used to regulate voltage without having any magnetic component in the circuit, it only uses a capacitor for energy storage as shown in Fig. 3(e) [15,16]. The capacitor is used instead of an inductor due to its advantageous factors like light-weight, high power density, extended voltage gains and low voltage stress but it faces drawbacks such as fixed voltage conversion ratio. To change this ratio, we need to increase the number of switches, capacitors and diodes that will lead to employing more gate driver circuits which causes very high current spikes and high Electro-Magnetic Interference (EMI) during charging of capacitors from voltage source. Its output power range is low whereas its efficiency has improved but unsatisfactory [9,10]. To improve its efficiency and reduced current spikes many switching methods are proposed in [17] for the low power devices.

2.1.2.2. Cascaded converter. When two buck-boost converters are cascaded, by adjusting duty cycle($0 < d \leqslant 1$) they are interconnected sequentially to provide regulated voltage output. In Fig. 3(f) cascaded non-isolated bidirectional converter is shown which is commonly used in energy storage systems. It uses only one inductor in the circuit due to which it is small in size and has high efficiency. These converters are operated at a higher power rating whereas ripple in current and stress is reduced of inductor, diode and switch stress are reduced. Also, high voltage gains at output's end [10,11].

2.1.2.3. Interleaved converter. For higher power applications conventional boost converter is employed but it is having various limitations such as high current and switching frequency ripples, voltage stress, reduced power density and Electro-magnetic interference; to overcome these issues interleaved bidirectional converter as shown in Fig. 3(g) is employed which is having interleaved techniques to reduce ripples in current and switching, minimized Electro-Magnetic Interference(EMI) and it also improves the power factor [18,12]. Interleaved Boost

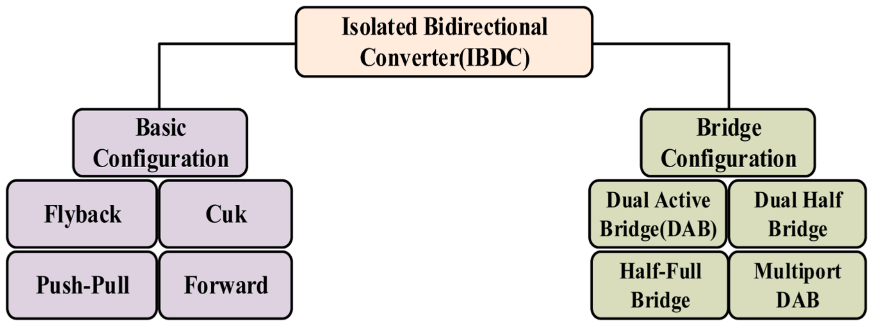

Fig. 5. General classification of Isolated Bidirectional topologies.

Converter is proposed for the MPPT Based SPV System to enhance its output using P & O method for maximum extraction [19].

2.1.2.4. Multilevel converter. Voltage output of the buck-boost converter acts as voltage input for multilevel converter as shown in Fig. 3 (h), output voltage is relatively high. Since the semiconductor switches counts are more which increases current ripple. To reduce them, extra capacitors and switches are employed in [13,20] for medium power applications.

2.2. Isolated bidirectional converter

Isolated Bi-directional DC Converter (IBDC) transfers power using High-Frequency Transformer (HFT) which is advantageous over Non-Isolated Bi-Directional Converters (NBDCs) in terms of high voltage gain ratio, wide input range. Also, galvanic isolation is provided into a transformer which causes magnetic interference and it also increases its weight, as the frequency increases switching loss also increases. Bi-directional converters are proposed in [21] for the interaction of Electrical Vehicles (EVs) and Grid. IBDCs fulfil load demand using galvanic isolation into the converter with the Energy Storage System (ESS) [22]. IBDC in [23] is for the improved operating range of power flow characterization. In bi-directional converter is termed as the core of next-generation medium voltage power conversion it states that in upcoming time 3.3 kV/6.6 kV converter of silicon carbide and gallium nitride will be available in the market [24]. For automobile application, which trails to boost efficiency from 89.6 % to 93.5 % a dual active bridge is proposed [25]. In [26] bi-directional dc converter is used with flywheel for ultra-fast on route charging system which provides 50 % of energy during on route charging and also reduce peak demand up to 66 % based on case study. Further, these converters are classified into basic configuration and bridges as shown in Fig. 5. These basic configuration IBDCs are comparable to the different types of basic NBDCs whereas bridge topologies are resultant by the changing unidirectional switches of different bridges with bidirectional switches.

2.2.1. Basic topology

2.2.1.1. Flyback isolated bidirectional converter. A flyback is analogous to non-isolated buck-boost DC-DC converter, but isolation transformer is employed instead of an inductor as shown in Fig. 6(a). It stores energy during the ON state and delivers it to load during the OFF state but during OFF state it experiences ringing and high voltage spikes at higher frequencies. It is employed for low power applications. The major drawback of flyback converter is its leakage inductance, flyback converter with a snubber circuit is proposed in [27] for suppressing transient voltage and dissipating its energy into the associated resistor. A ripple cancelling circuit is proposed for the elimination of ripple into the input current having a transformer, two capacitors and winding are employed to the core of a conventional flyback converter transformer which behaves as a snubber circuit it reduces 6 % current ripple with minimum power loss [28]. A single-stage step-up converter with galvanic isolation is discussed in [29]. Which consists of two converter a push-pull (PP) converter and a flyback converter in which first performs the function of a DC transformer and the second as voltage stabilizer. Input voltage, output voltage & power range are 24V-32 V, 400 V, 100W-400 W and 1 MHz under ZVS respectively. It achieves peak efficiency of 97.1 %. over a wide voltage range. In [30] a prototype flyback converter is designed and has an efficiency of 92.5 % is proposed for energy collection based on Thermo Electric Generator (TEG) by rearrangement of the Flyback converter so that its partial energy is processed and the rest is directly delivered to the load, this causes reduced current and voltage stress also increases its efficiency. In [31] multiple outputs are shown of flyback converter by joining multiple winding SMPS transformer in circuit with two part one master and other is slave. Master is achieved by connecting any output directly to the feedback control loop and slave is obtained by tuning turn ratio of tertiary winding. But this set-up led to inductance leakage in between secondary and tertiary winding which causes a reduction in accuracy of output voltage; to overcome main output is provided with its feedback. [32] proposed isolated dc converter with buck-boost and forward flyback. It is advantageous due to its extended range of voltage, increased voltage gain and continuous current characteristics with leakage inductance recovery scheme by combining LV side inductor with the transformer, by implementing its result are verified and its efficiency dives to 96.9 % and 94.5 % in buck & boost mode respectively. An interleaved flyback converter is proposed with parallelly connected input of N modules with single output for a photovoltaic generation system. Also, single core transformer is employed with N primary winding and single secondary winding which is advantageous over conventional interleaved input parallel output series topology. In this proposed converter, number of flyback modules are equal to the individual core of the transformer. In a cycle for this proposed structure output diode operates more N times that results to smaller value of output capacitance, a snubber circuit is also employed to avoid leakage inductance and *dv/dt* protection during commutation [33]. A flyback converter is employed for low power application in [34] it is applied for LEDs using partial energy processing in which the conventional flyback converter is rearranged in such a manner LED is placed in parallel with secondary and in series with a voltage source on primary, this provides partial processing of energy that means partial energy is directly delivered to load & rest is processed by converter. The proposed converter is having improved efficiency comparably to conventional flyback and less stress due to rearrangement of the flyback converter but the major disadvantage of this converter is not having any magnetic isolation and its power factor is dependent on the gain. Hybrid converter (buck-boost/flyback) is also proposed for application of an electronic sign indicating LEDs with 200 W, for battery charging and discharging. It is operated with Zero Voltage Switching (ZVS)which improves switching loss and it is advantageous over traditional because of its lightweight, compact size and improved efficiency by 4 %; while operating at full load its efficiency dives up to 85 % and its prototype is also designed with solar PV as input which delivers output

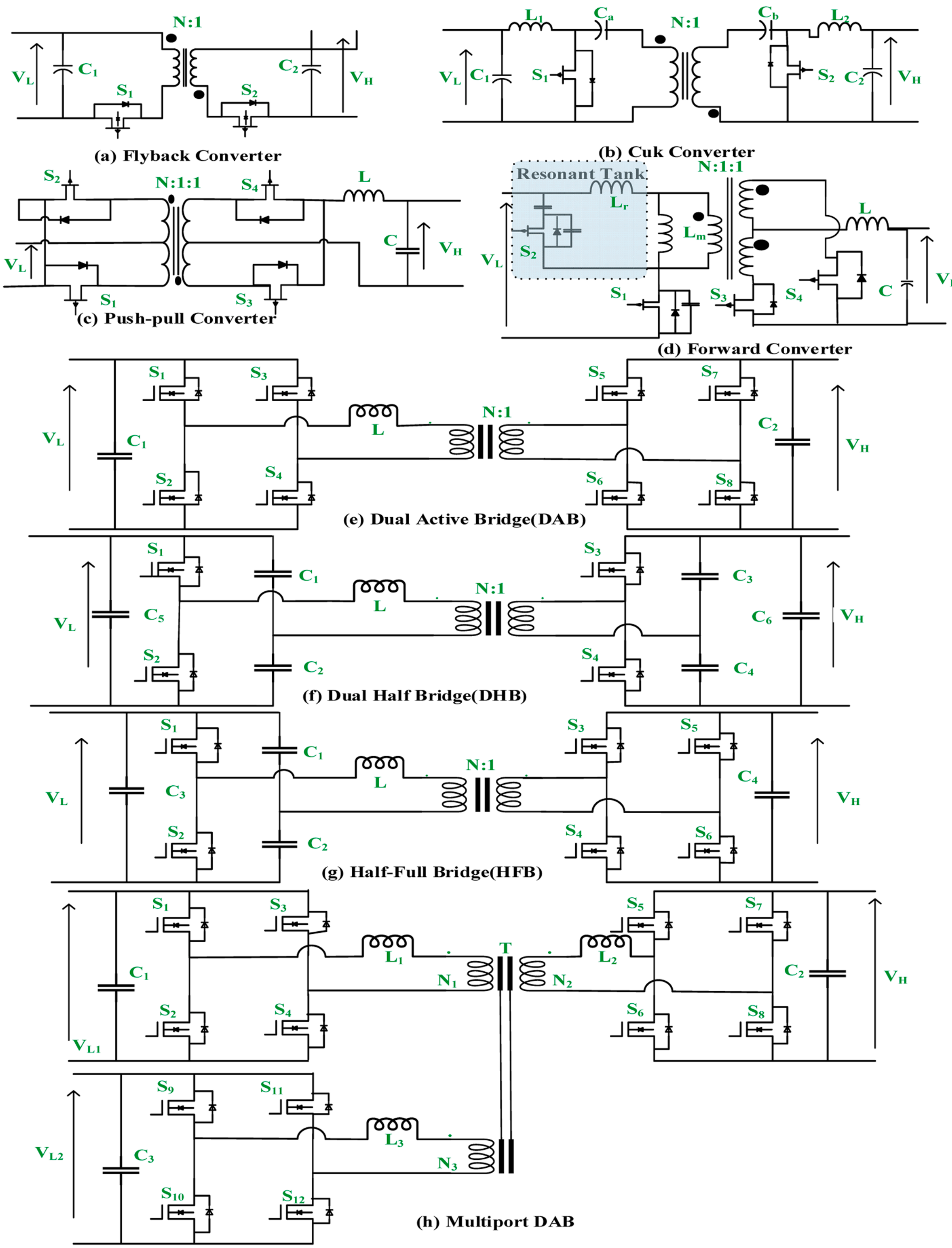

Fig. 6. Basic and bridge topology of IBDC.

voltage of 10 V and 20 W maximum output voltage [35]. Application of different proposed flyback converter illustrated in Table 3 which comprises input, output voltage with its efficiency. An interleaved flyback is proposed in which parallelly connected converters on both ends and termed as converter-1 and converter-2 the proposed converter is operated under SPWM with peak current control method [36]. A battery charger is proposed [37] in which diodes of flyback converter are replaced with MOSFET and joined with a High-Frequency Transformer (HFT) which offers high voltage gain, galvanic isolation for battery DC bus also provides protection from faults. A novel IBDC is proposed for numerous applications like HEVs, energy storage systems, fuel cell vehicles and grid interfacing. In this proposed converter voltage gain is enhanced due to a coupled inductor which behaves as a forward and flyback converter; soft switching is achieved by utilizing leakage inductance of the coupled inductor. 1 kW prototype is designed and implemented which shows peak efficiencies of 95.4 % and 93.6 % in buck and boost operation respectively [38]. Higher self-capacitance in a Flyback converter is produced when additional turns on secondary

Table 3
Various application and efficiency of flyback converter with its input, output voltage rating.

Reference	Input Voltage	Output Voltage	Efficiency	Applications
[27]	96V	590V	95 %	Electric Traction, DC supply
[29]	24–32V	400V	96.4 %	DC Microgrids, Satellites communication
[30]	21V	30.5V	92.65 %	Aerospace Industry, Battery chargers
[32]	24–55V	400V	94.5 % - 96.9 %	Industrial Process, Batteries of EVs
[38]	48V	400V	93.6 %- 95.4 %	Distributed Generation system

Table 4
Various application and efficiency of Cuk converter its input and output voltage rating.

Reference	Input Voltage	Output Voltage	Efficiency	Applications
[41]	30V	90V	97 %	Electric Vehicles, Hybrid System
[42]	160–260V	12V	91.15 %	EV Chargers, Home appliances and small power tools
[43]	12V	48V	92 %	Integration between low voltage high current source and energy storage devices

winding is present. In [39] flyback converter for R load is modelled also studied of parasitic capacitances and found a leading impact of the parasitic capacitor into converter, conventionally it is ignored into ideal flyback converter. For energy balancing, two flyback converter is discussed for a battery pack with series connection which cause energy transfer between battery or a single cell. It also decreases components of energy storage which led to reduction in its volume and cost for balancing system, its balancing speed is also improved by proposed topology i.e., Dual Objective hybrid control strategy. [40].

2.2.1.2. Cuk isolated bidirectional converter. Conventional Cuk converter consists of capacitor and inductors, each inductor is placed at input and output for boost and buck topology respectively. It stores energy into the inductor during the first operating mode and loses it to load during the next operating mode. Fig. 6(b) shows cuk converter which can be incorporated with isolation between equally split capacitors into the primary and secondary segments. For high-power applications, frequency and transformer turn ratio has to be high which helps in achieving high voltage gain but it also causes high leakage inductance too. To minimize leakage inductance many methods can be used like soft switching, snubber circuits etc. In [41] Cuk based IBDC with resonance for EV application is designed and simulated which increases its efficiency compared to the circuit without an active snubber, it achieves soft switching and is controlled by a dual-loop method. Cuk converter is employed in EVs, Hybrid systems etc. In [42] Cuk converter is designed with a corrected power factor for application in home as a brushless dc ceiling fan, it operates in continuous conduction mode which ensures approx. unity power factor with a low torque ripple of 2.14 % whereas at full load it is 4.72 % with the power factor of 0.999 which makes it suitable for home appliances and small power tools. It is also used for the automotive dc source which can be further used for the interfacing between the DC link and batteries. In [43] an isolated bidirectional Cuk converter is designed for the interfacing between the energy storage device and low voltage high current source, it operates at a low voltage and high current. Also, a prototype is also designed of rating 1.2 kW for the verification of the system in which input voltage range is 1.5 V to 6 V, the maximum input current is 300A (from 2 V to 4 V) and the output voltage is 24 V. Major issues that arises while using Cuk converter in motor drives are harmonic distortion and uneven output voltage. Isolated Cuk Converter is proposed with two symmetrical output voltages for SRM drive with high-frequency isolation is provided for reduced input current THD and regulating dc output voltages; which reduces system size, cost and its complexity whereas it limits input current THD within 5 % and shows smooth transient operation with unity power factor under different load variation when it is operated under DCM [44]. Application of different proposed Cuk converter is illustrated in Table 4 which comprises input, output voltage with its efficiency.

2.2.1.3. Push-pull isolated bidirectional converter. Two forward converters are connected in cascaded manner to form push-pull converter which operate back-to-back, a centrally tapped transformer where the secondary side consists of a buck converter. Centrally tapped winding behaves as a simple forward converter that operates on an alternate cycle, to utilize the core it magnetizes in both directions. Fig. 6(c) shows a push-pull converter unlike flyback and other topologies. For power transformer it utilizes transformer action from primary to secondary. Input and output of transformer is regulated in a feed forward arrangement calculated by transformer turns ratio; it shows if input voltage is regulated then output voltage is the same [45]. It is utilized with resonant voltage doubler rectifier which provides higher voltage gain and reduced turns ratio of a transformer, employed in standalone PV system with battery energy storage [46]. An active clamping circuit is utilized on the low voltage(inductor) and for high voltage(capacitor) is to achieve zero voltage switching [47]. Push-pull is employed in a multifunctional isolated microinverter that injects power into the power grid by utilizing the maximum available solar PV module by conversion from DC-DC to DC-AC simultaneously. It is also utilized for charging integrated energy storage system without any extra circuitry except the addition of one diode and a capacitor; while employing it into PV system major issue faced is not maximum utilization of available solar energy to overcome these issues many algorithms are proposed in [48–50]. A push-pull converter is employed into a PV microinverter without using any electrolytic capacitor and provides input to the current sensor for overcoming issues of 2nd harmonic in frequency. DC bus which creates major side effects like narrow frequency bandwidth, harmonic voltage ripple and limited PV voltage control loop and input power calculation [51]. Major issue with the converter is discontinuous value input current which is very much required while integrating with renewable energy sources. To overcome, author of [52] proposed a converter with four stages of conversion as impedance network, switching, isolated transformer and voltage multiplier rectifier are first, second, third and fourth stage respectively; by using continuous current is observed, its size and weight are reduced and also 260 W prototype is designed & analyzed which delivers its efficiency can be above 91 %. Many characteristics are explored to improve efficiency and decrease switching losses, in addition new converters are proposed with fewer switches. Different voltage gain transformer turns ratio and duty cycle are adjusted but while in comparison with conventional converter and found they are having same level of voltage stress [53]. A three winding isolated push-pull converter is employed for higher efficiency over wide input voltage range by commuting primary with ZVS and secondary by ZCS for verification. A prototype of 280 W, switching frequency of 34.7kHz with input and output voltage is 10 V and 280 V respectively attained maximum efficiency of 95.4 % whereas output power is 121 W [54]. A controller mapping for bidirectional modulated push-pull converter is preferred to operate in optimal zone that enhances input voltage range and it is proposed in three-level PWM with phase shift control scheme, a

Table 5
Various application and efficiency of push-pull converter its input, output voltage rating.

Reference	Input Voltage	Output Voltage	Efficiency	Applications
[52]	30V	240V	< 91 % over the whole load range	Photovoltaic arrays, Fuel Cells
[54]	10V	280V	95.4 %	DC Drives
[55]	30–48V	380V	94.1 % peak efficiency (boost mode) 94.5 % peak efficiency (buck mode)	DC motor drives, Distributed Renewable Energy System
[56]	40–59V	380V	95.8 % peak efficiency 93.8 % efficiency at half load condition	UPS, Storage devices
[57]	60V	300V	95 % peak efficiency 93 % efficiency at half load condition	Energy Storage System
[58]	48–72V	380V	96.85 % forward direction 96.44 % backward direction	Battery Industries
[52]	30V	240V	< 91 % over the whole load range	Photovoltaic arrays, Fuel Cells

prototype is verified with analysis which operates under 12 operating modes with the expanded soft-switching range with input and output voltage of 30–48 V and 380 V respectively [55]. The author of [56] also proposed a converter in which natural commutation is provided on low voltage side with ZCS and ZVS on higher voltage side which eliminates extra active and passive clamp circuits since higher frequency is used which reduces volume of transformer which makes it perfect for high power application. Also, an experimental setup of 1kW is established to validate result peak efficiency is observed as 95.8 % with the input voltage of 59 V. In [57] zero-voltage switching (ZVS) converter is proposed in which one more switch is added to primary side to achieve ZVS whereas on secondary full bridge topology is used with driving circuit which shows improvement in stability of whole system, it is shown maximum efficiency of 95 % but it decreases to 93 % at half load condition with input voltage of 60 V and output voltage is observed as 300 V. A converter is derived by integrating a bidirectional switch with PWM current-fed push-pull converter with an active voltage doubler which operates in forward direction but a half-bridge PWM converter operates in backward direction it achieves ZVS over wide range of battery voltage where its input voltage is 48–72 V and its voltage across secondary side is 380 V which shows maximum efficiency of 96.85 % in forward direction and 96.44 % in backward direction [58]. While employing push-pull converter in various systems few authors have found push-pull can be employed for battery charger & for interfacing between battery and grid using DC bus at the time of grid failure even it can be used for the wireless power transfer system [59,60]. Table 5 illustrates the application of various proposed push-pull converters, including input and output voltage and efficiency.

2.2.1.4. Forward isolated bidirectional converter. Forward converter as shown in Fig. 6(d) transfers energy directly without storing energy. During ON state, it utilizes an AC transformer without an air gap instead of coupled inductor. While ON state input voltage causes increased magnetizing current but during OFF state it decreases, its voltage second balance is maintained by its magnetizing current reaches zero before switching period ends. For controlling a clamp circuit ZVS can be achieved for soft switching whereas it is applied in a range of 200W- 500 W with single and double switch respectively. Based on requirements and output hybrid configuration of the converter is proposed in various literature like push-pull-forward converter, forward-flyback converter, interleaved-forward converter. A hybrid configuration of forward and flyback converter with recycling of leakage energy is designed on a prototype circuit of 200 V/24 V with output power and switching frequency of 500 W and 50kHz respectively. It is implemented with two transformers for the reduction of current stress, conduction loss and to improve its efficiency; it operates in two modes. Step-up involves clamping voltage through switches at input and recycled energy is sent back to the voltage source. Energy lost during leakage is recycled to the clamping capacitor because switch voltage is clamped from input during step-down [61]. A single-stage forward flyback power factor correction converter is proposed by merging flyback and forward converter with a common transformer that operates by sharing the output power in the rest region and it is controlled by quasi-resonant. Two different quasi-resonant (QR) have been studied and the prototype is designed to verify results in [62]. A resonant forward-flyback converter with suppressed frequency variation shares its transformer and primary switches in which soft switching is achieved for primary switches in forwarding mode and flyback mode, it resets the transformer by transferring energy to output [63]. An interleaved two-switch forward converter is proposed with zero voltage switching and zero current switching with a coupled inductor and transformer leakage inductance, snubber and parallel capacitance which is advantageous over another converter having coupled inductor which decreases voltage stress and parasitic ringing [64]. Aim to implement any topology or form a new converter is only to reduce snubber elements and the volume of the transformer. A new converter with zero voltage switching and resonant core reset PWM forward converter is introduced in [65] with an auxiliary circuit with a low no. of switches that operate under soft-switching with an overall efficiency of 93.5 %. A novel active-clamp forward converter (ACFC) is proposed in comparison to a typical active clamp forward converter, lowers voltage spikes on free-wheeling and forward rectifier diodes by using a lossless snubber on the secondary side that consists of a resonant capacitor, clamping diode and output inductor in parallel [66]. A new converter arrangement is proposed in which two identical switch forward converters shares transformer which reduce voltage stress of power switches by voltage clamper as well as recycles leakage energy of the source. Prototype is designed to validate the proposed converter in which its input voltage, output voltage, switching frequency and the maximum output voltage is 12 V, 48 V, 100kHz and 20 W respectively. The resonant version of this converter is proposed by utilizing a transformer leakage inductor as a resonant inductor. A new forward type resonant bidirectional dc-dc converter is proposed in which zero current switching is achieved without engaging any additional circuitry; its switching losses are reduced which shows it can be operated with higher frequencies for low and medium power applications [67,68]. When two forward converters are joined with a dual transformer, to design this converter a resonant inductor and two switches are required. For reduced switching loss zero voltage switching is achieved by employing a leakage transformer and by paralleling the transformer secondaries we can share loads evenly among inductors which reduces current stress, conduction losses [69]. With low dc offset current for transformer, double-ended active clamp converter is proposed primarily for digital devices supply. But after employing an extra diode while turn-off clamp capacitor gets charged proposed converter can reach lower dc offset voltage and reduce conduction losses for medium and heavy loads [70]. Cascading is also a solution but [71] suggested a new method of the boost-type resonant forward converter with a topology combination to increase voltage by using switch and diode simultaneously resonant power conversion respectively this converter emphasized the performance feature of two stages forward converter which has improved high power efficiency due to compact power conversion stages. A bidirectional buck-boost converter and forward-flyback converter are

Table 6
Various application and efficiency of forward converter its input and output voltage rating.

Reference	Input Voltage	Output Voltage	Efficiency	Applications
[61]	24V	200V	92.7 % (Step up) 95.5 % (Step down)	High Power applications
[62]	90–265V	80V	92.2 % Peak efficiency	LED drivers and products
[65]	85–180V	24V	93.5 %	LED Adapter, Voltage regulators
[67]	12V	48V	94 % (Step up) 92.9 % (Step down)	Drivers circuit
[68]	48V	200V	92 %	Low & medium power applications
[69]	130–230V	24V	92.5 % Maximum Efficiency	Low power applications
[70]	400V	12V	92.8 % Maximum Efficiency	Power supply for digital devices
[32]	400V	24–55V	96.9 % (Step-up) 94.5 % (Step down)	Electric Scooters

composed to achieve enhanced gain and improved input voltage range by utilizing continuous current characteristics at the low side port during step up and down mode. It has features of leakage inductance energy recovery; switches are having reduced voltage stress and ZVS is achieved it roots improvement in efficiency [32]. Proposed forward converter with different application is shown in Table 6 which comprises the input, output voltage with its efficiency.

2.2.2. Bridges topology

2.2.2.1. Dual active bridge (DAB). The bidirectional isolated Dual Active Bridge (DAB) dc-dc converter comprises two full-bridge converters with galvanically isolated high-frequency transformer which allows to affords small sized converter. DAB is chosen for its high voltage-conversion ratio, higher efficiency and wide ranged voltages which permits the interface of low-voltage solar modules with a high-voltage dc bus, such as the input of a micro-inverter [72]. As shown in Fig. 6(e) DAB is symmetrical in a circuit which is having two voltage sides with the unity transformers turns ratio i.e. N:1, the operation of DAB is characterized by duty ratios and phase shift between the AC wave and capacitor allowing to achieve zero voltage switching(ZVS) it is widely used from the range of few kilowatts to several hundreds of kilowatts. Dual Active Bridge (DAB) is advantageous over other bridge converters such as lightweight, low volume and cost; it also avoids distortion between current and voltage. When the rating of switches is the same then transmission power is decided by the no. of switches; more the switches more power is transmitted which makes DAB's biggest power capacity converter but it has some disadvantages of high switching loss when it is employed for light loads due to inadequacy in ZVS whereas for heavy loads it faces high conduction losses due to high circulating current [73]. Authors mainly focus on expanding their rating, characteristics, control strategy, power density, electromagnetic noise, efficiency and soft switching. For expansion, the voltage and current rating of DAB author of [74] suggested three possible arrangements by positioning two identical DABs in series and/or parallel arrangement. A hybrid input series output series converter (ISOS) constituted by resonant and non-resonant dual active bridge modules and input parallel output series (IPOS) modular dual active bridge converter with voltage balancing on hierarchical sliding mode control are proposed for application in dc grids and distribution power systems respectively [75,76]. Two modulation techniques are used to design a center-tapped resonant Dual Active Bridge (DAB) prototype into which it blocks reverse current also eliminates back power flow for moderated transfer losses [77]. Ideally, ZVS is achieved when the current flows through its anti-parallel diode before commuting to the device but when ZVS is not achieved it causes ringing into the voltage waveform triggered by sudden discharging of parasitic capacitance and semiconductor devices produce losses, to avoid ZVS limits and AC currents for Dual Active Bridge is developed providing phase shift and different duty cycle [78]. To obtain high voltage conversion ratio, galvanic isolation high frequency (HF) transformers are employed but it produces a dc bias in the transformer which is further classified into transient and steady dc bias; to eliminate conventionally flux measuring and balancing methods are used by implementing prediction and suppression method [79,80]. To suppress DC bias voltage, traditionally a DC blocking capacitor was used but it changes system impedance. To overcome this issue many modified methods have been proposed by researchers a modified modulation scheme is proposed to reduce the transient DC bias current by staggering the second value of primary and secondary bridges [81]. A medium frequency transformer is modelled and designed for the improvement of overall power density by integrating a phase-shifting inductor (PSI) in the transformer; keeping ecological aspects an eco-dimensioning approach is given for planer transformer in DAB application which evaluates converter energy consumption over the complete converter lifetime [82,83]. During transient achieving constant output voltage with precised result of DAB is tedious task and for achieving a fast response various feed-forward control strategy has been proposed, a single-phase shift control scheme(virtual direct power) is proposed to improve results during start-up, load variation causes no overshoot and also provides fast transient response of converter [84]. Voltage feeds a dual active bridge which would manage high peak and huge RMS current that causes a very large current ripple which affects system efficiency and reliability; to overcome these issues, high efficiency CF-DAB converter over wide load range with optimized a switching pattern which is controlled by a combination of PWM duty cycles and phase shift to achieve zero voltage switching throughout wider range of load [85]. A proposed Current Fed Dual Active Bridge (CF-DAB) with dual transformer for interfacing EVs and Energy Storage System (ESS) for wider voltage range, an optimal switching pattern is

Table 7
Various applications and efficiency of DAB converter its input and output rating.

Reference	Input/Output Voltage	Power Rating	Turns Ratio	Switching Frequency	Efficiency	Application
[72]	220	85W	1:13	15kHz	93.1 %	Microinverters
[74]	750Vdc	100kW	1:1	20kHz	98 %	DC grids
[75]	150 V/150V	600W	1:1	10kHz	94.9 %	Distributed Power System
[85]	18–28 V/300V	1kW	2:10	50kHz	Buck Mode-97 % Boost Mode-97.2 %	Electric Vehicles, Energy Storage System
[86]	18–36 V/250–400V	1kW	1:5	80kHz	(For 36 V) Buck Mode-95.1 % Boost Mode-95.2 %	Electric Vehicles

proposed to reduce leakage inductor peak and by controlling all variables current stress, conduction loss is reduced [86]. An algorithm is provided for DAB in [87] to meet industry requirements using current filters and DC-bias rejection processes by changing power flow in transient; converter input and output will be the same as steady-state operation. The application of dual active bridges is increasing day by day in photovoltaic systems, plug-in electric vehicles, battery chargers, the future aviation industry and many more industries: to employ DAB different designs and methods are proposed, a unified voltage balance control is simulated for direct utility interfacing of charger/discharger by using cascaded H-bridge and current fed dual active bridge in which cascaded H-bridge is controlled by unified voltage balance whereas DAB is current fed [88]. Dual Active Bridge is employed in the aircraft industry for improved efficiency, power density. Aircraft HVDC (+/-270VDC) is facilitated for the distribution it caters significant flexibility for power transfer, ratings and inputs/outputs ports; which is interfaced with 28 V DC bus [89]. Authors of [90] implemented 3 methods to increase ZVS range for DAB in which first is to modify I_2 when ZVS is close to being lost second is to Turn off DAB for low power application and last is modification of inductor or switching frequency. Table 7 is included which shows different types of proposed Dual Active Bridge (DAB) converter with its input, output voltage, power rating, turns ratio, switching frequency and efficiency.

2.2.2.2. Dual half bridge (DHB). The Dual Half Bridge (DHB) is obtained by connecting HF transformer in between two half bridges as shown in Fig. 6(f). Dual Half-Bridge (DHB) are effective to improve the system performance, it is advantageous over full-bridge due to the wide ZVS range, reducing conduction loss; it has also minimum circulating current and continuous power transfer. Half-bridge topologies are employed where bidirectional power flow capabilities are required but on light load dual half-bridge faces high circulation current losses and high switching losses which makes it suitable for application up to 500 W; to overcome resonant converters are used with frequency modulation but this circuit has asymmetrical power flow characteristics in forward and reverse power flow directions. For symmetric power flow various circuit topologies are given like Zero Voltage Switching (ZVS) also resonant tank is employed in series with high voltage side and synchronous rectifiers on lower side to accomplish soft switching and reduced conduction losses respectively and during forward power flow parallelly placed inductor is disconnected results reduced circulating current loss [91,92]. Split capacitors in Dual Half Bridge (DHB) improves power flow capability, efficiency and to reduce switching losses, soft switching characteristics is achieved by adding frequency modulated series resonant tank at high-voltage side [93]. A DHB with hybrid rectifier then compared to centrally tapped full bridge rectifier; a controller for capacitor less current fed dual half-bridge converter is proposed for the grid-connected fuel cell to increase the lifetime of the converter and achieve ZVS dc link is suppressed, the proposed controller reduces input current ripple and dc-link voltage [94,95]. Converter with hybrid half and the full Dual Active Bridge (DAB) is recommended for matching voltage, wide range and voltage conversion ratio by adding two switches and two capacitors [96]. Voltage gain can be enhanced by adding the output of two dual half-bridge into series with a voltage multiplier at the output for the improved voltage conversion ratio which led to reduces voltage stress across switches whereas voltage ringing is suppressed by adopting an active clamp circuit as discussed in [97]; Two parallelly connected half-bridge converter proposed in [98] with additional inductor to the primary of converter for energy storage which led it to operate under zero voltage switching, power transfer during whole time period and removes output filters since no circulating current is present these advantages makes converter fit for application in high power, voltage and flexible inputs. One of the key factors for efficiency of power switches are its materials; a comparative analysis is performed for a half-bridge LLC resonant converter based on material silicon carbide (SiC) and silicon (Si) also it is observed silicon carbide (SiC) is more effective comparably silicon (Si) but considering cost as an important factor silicon (Si) is used more comparably silicon carbide (SiC) for the high frequency applications [99]. When a half-bridge DC-DC converter and a quasi-switched capacitor converter are combined, soft switching is achieved and voltage stress is also decreased by one-fifth of the output voltage in primary and secondary sides respectively resulting output efficiency of 91 % at 5 kHz [100]. When current fed dual half-bridge converter is directly connected with the half-bridge inverter for fulling the power demand of the residential requirements, it faces the issue of voltage fluctuation and voltage imbalance which is tackled by employing a repetitive controller [101]. A novel zero voltage switching (ZVS) is proposed which comprises two half-bridge converters in parallel arrangement on the primary side and a current doubler rectifier on the secondary side with two output filter inductors operated by phase-shift control; compared with the traditional phase-shift full-bridge converter proposed converter is having wide ZVS range with minimized ripples and circulating current of output filters [102]. A dual half-bridge converter is employed in various systems like DC power supply in railways applications for interior lighting, battery chargers, electric vehicles, an interface between two DC-link voltage sources, and residential photovoltaic systems [101,103,104]. Table 8 is included which shows the different type of proposed DHBs with input & output voltage, power rating, turns ratio, switching frequency and efficiency.

2.2.2.3. Half-Full bridge. Half-Full Bridge (HFB) is derived by combination of half and full bridge arranged as primary side is HB, secondary is FB connected with high-frequency transformer as shown in Fig. 6(g). A current controller is designed for HFB converter that is employed for balancing link of battery; the half-bridge is connected to the primary whereas full-bridge is connected on secondary both are interfaced by a high-frequency isolation transformer which is responsible for energy transfer when input filter capacitors are replaced with two battery cell at half- bridge side, now this proposed converter is capable of battery balancing, high and low voltage conversion [105]. Major issues for High Voltage DC(HVDC) transmission system are short-circuited fault and capacitor voltage imbalance; a new topology is simulated of the half-full bridge is proposed in which conducting diode is forced off whereas capacitor voltage balancing is accomplished using charging and discharging of two capacitors at same time in absence of any complex balancing circuit [106]. In [107] paper presents the development and verification of a model of DC-DC resonant converter used in fuel cell applications in electric vehicles, along with the development of transfer functions and the implementation of a PI controller to improve converter performance and stability. A hybrid converter is designed for the solution of wide voltage variation issue from solar photovoltaic panels due to variating sunlight; a half-/full resonant converter has equal resonant frequency both primary and secondary to attain bidirectional power flow by using a pulse frequency modulation approach [108]. Previously proposed models are having same dynamics because it is assumed that their submodules capacitor voltage are well balanced but when the negative voltage in hybrid multi-modular converters is taken into account, the dynamics of HB-SMs and FB-SMs change.; in [109] an efficient and accurate model is simulated it is assumed as capacitor voltage are well balanced, switching events are neglected and it is compared on parameters like- power flow, faults, numerical calculation and analysis are performed in [110]. A phase-shifted carried (PSC) technique with a boosted modulation index and capacitor sorting algorithm is proposed to resolve issue for asymmetric voltage reference wave of half-full hybrid modular multilevel converter (MMC); it improves voltage balance and also reduces harmonic distortion [111]. Existing balancing strategies and battery balancer with auxiliary power module is proposed in [112]. Since indirect power (Pind) directly affects the volume and losses of passive components, a thorough study and comparison based on the Half/Full Bridge (HFB) and Interconnected

Modular Multi-level Converter (IMMC) topologies, comparison is performed on various parameters like circuit specification, approximation (ripples, parasitic, Quasi-static analysis), voltage and current waveform (capacitor, output) when compared to a half-full bridge arrangement, IMMC processes roughly twice as much power in passive components [113]. A comparison is performed by using different power switches for Modular Multilevel Converter (MMC) with half/full bridge and found conduction, switching losses of capacitor in semi-full bridge for MMC is reduced as compared to half-/full-bridge converter MMC by 1.8 % and 1.7 % respectively [114].

2.2.2.4. Multiport dual active bridge (DAB). During the integration of multiple input voltage sources and hybrid electric vehicles, multi-input converters are preferred Fig. 6(h) shows multiport DAB consisting DC-link and magnetic coupling, which allows integration without using extra additional circuits or switches. Issues are the same whether it is with grid interfacing or standalone system, solar photovoltaic systems with batteries are very common arrangements, the multiport converter is advantageous to conventional structures with minimized conversion steps, lower component count, higher efficiency and easy centralized control. Interfacing low and high voltage DC bus efficiently is all times difficult; integrating dc bus with hybrid storage system Current Fed Dual Active Bridge (CF-DAB) with high gain multiport dc converter with three-port interface, it operates under ZVS region voltage multiplier cells are used to obtain high voltage conversion ratio between super-capacitor and dc bus [115]. 6 topology of three-port converters are given in different combination of DC linked, magnetically coupled and current-fed multiport bidirectional DC-DC converters [116]. A converter is proposed and designed to reduce stages and average switching frequency using a modified modulation scheme by employing CF-DAB which offers galvanic isolation and a high voltage conversion ratio [117]. This arrangement allows us to interconnect multiple sources without adding any additional switches which makes this topology simpler and has a minimum number of power devices this results in lower switching losses and less electromagnetic interference [118]. [119] Proposed Hill-climbing technique-based controller which allows for three step DC-DC converter for fulfilling load demand. A four-port bidirectional dual active bridge converter is designed for the EV fast charging in which the first, second, third and fourth port is used as source, load, source and load both respectively; when these ports are used to supply power, it acts as source but when it is used to charge the battery then acts as load [120] these results to lower the switching losses and less electromagnetic interference. A five-level cascaded bridge is introduced with less harmonics for better performance using GA algorithm [121].When multiport converters are integrated the practical stability of the system is also a major issue, to overcome this a single-phase solid-state transformer is designed for electromagnetic timescale for every subsystem which is fed back with closed-loop transfer function which requires two current sensors and sinusoidal perturbation current source. MMC and cascaded bridges are utilized to create a multiport converter its description is divided into three cases with combination of multilevel converter port, cascaded bridges and different nos. of output ports to check its feasibility a prototype is designed in [122]. The multiport dual active bridge is used to interface additional elements like batteries for storage; a design is developed for upcoming aircrafts to employ power scalable BDC [123]. A three-phase 3-port dual active bridge converter is proposed in wye-wye-delta arrangement with three-port, leading to reduced size and improvement in system reliability by manipulating parasitic capacitance and leakage inductance of the transformer [124]. A split dc-link dual active

Table 8
Various applications and efficiency of DHB converter its input and output rating.

Reference	Input/ Output Voltage	Power Rating	Turns Ratio	Switching Frequency	Efficiency	Applications
[92]	400 V/52V	480W	1:7.7	Forward Mode-99 kHz Reverse Mode- 71kHz	At 100 % load Forward Mode-93.6 % Reverse Mode- 91.2 %	Low Power applications
[94]	380–450 V/120V	1.2kW	$0.5(1 + D)$	100kHz	For 380 V- 96 % For 450 V- 95 %	Grid Connected Fuel cell system
[96]	120–240 V/96V	1kW	10:4	50kHz	95.1 %	Distributed Power System
[97]	15–30 V/400V	1kW	1:15	100kHz	97 %	Electric Vehicles
[99]	300 V/400V	3kW	1:1.5	100kHz	98.1 % at 300V	EVs, Aircrafts

Table 9
Various applications and efficiency of Multiport DAB converter its input and output rating.

Reference	Input Voltage	Output Voltage	Switching Frequency	Applications
[120]	$V_{1\&4} = 400V$	$V_{2\&3} = 600V$	50 kHz	Smart grids
[127]	220V	800V	19.2kHz	Microgrids
[123]	230–290V	28V	400Hz	Future Aircrafts
[124]	5500V	400V	10kHz	Medium voltage applications
[125]	48V	380V	15kHz	Microgrids, Hybrid PV and battery system

Table 10
Application, Power Rating and Comparison of IBDC Topologies.

Reference	Configuration	Topology	Winding	Inductors	Capacitors	Switches
[29,30]	Basic Configuration	Isolated Bi-Directional Flyback Converter	2	0	2	2
[41,42]		Isolated Cuk Converter	2	2	4	2
[52,55,58]		Isolated Push-pull Bidirectional converter	4	1	1	4
[62,65,70]		Forward Isolated Bidirectional Converter	3	1	1	3
[72,75,85]	Advance Configuration	Dual Active Bridge (DAB)	2	0	2	8
[92,96,101] [103,104]		Dual Half Bridge	2	0	6	4
[120,123] [124,125]		Multiport Dual Active Bridge	$n = 3$	0	$n = 3$	4n=12

Table 11
Advantage and Disadvantages of IBDCs.

Advantages	Disadvantages
1. Equal current switch stresses on both bridges	1. The converter may lose soft switching under light load condition.
2. Soft switching can be achieved easily.	2. High ripple into DC buses requires additional filtering circuits which makes circuit complex.
3. Simple construction	3. Extremely sensitive control is required when higher bus voltage is available.
4. Fast Dynamic Response	4. Higher no. of switches leads to greater volume, higher losses compared to low switch count.

bridge (SDLDAB) is designed for interfacing of PV modules and battery bank with dc microgrids; with favoring weather conditions it is operated at Maximum Power Point Tracking (MPPT) by maintaining the voltage at input terminals of PV modules, battery and dc microgrid are connected with a transformer to achieve high voltage gain, modulating switching sequence is used to minimize transformer current and direct power flow is achieved to battery without involving transformer [125]. A multilevel dc converter is proposed for fast charging of electric and plug-in hybrid electric vehicles which consists of full-bridge converter and voltage doubler rectifier units at the primary and secondary side respectively to generate the multilevel voltage [126]. Table 9 is included which shows various proposed Multiport Dual Active Bridge (MDAB) converters with input & output voltage, switching frequency and its applications.

3. Resonant converters

Many scholars have investigated various forms of DC-DC converters designed for the utilization of renewable energy and energy storage into EVs, with the aim of enhancing efficiency and addressing the limitations of these converters. DC-DC converters can be classified into three operational modes. These are linear mode, hard switching and soft switching mode. The linear mode is characterized by simplicity, low noise with satisfactory regulation, and rapid response on the other hand its disadvantage lies in its low efficiency attributes to power dissipation in diverse operational scenarios. Hard switching mode converters are classified into non-isolated and isolated converters, depending on the presence of galvanic isolation. The disadvantages of hard switching mode converters encompass elevated level of electromagnetic interference (EMI), substantial switching losses, and considerable dimensions and mass, all of which exert an impact on switching frequency. Last category is soft switching converters or resonant converters, it provides solution to the challenges in the hard switching as shown in different studies [128,129]. Its operation is categorized into two types i.e. zero current switching (ZCS) and zero voltage switching (ZVS); further Soft Switching based DC-DC converters/Resonant Converters are categorized

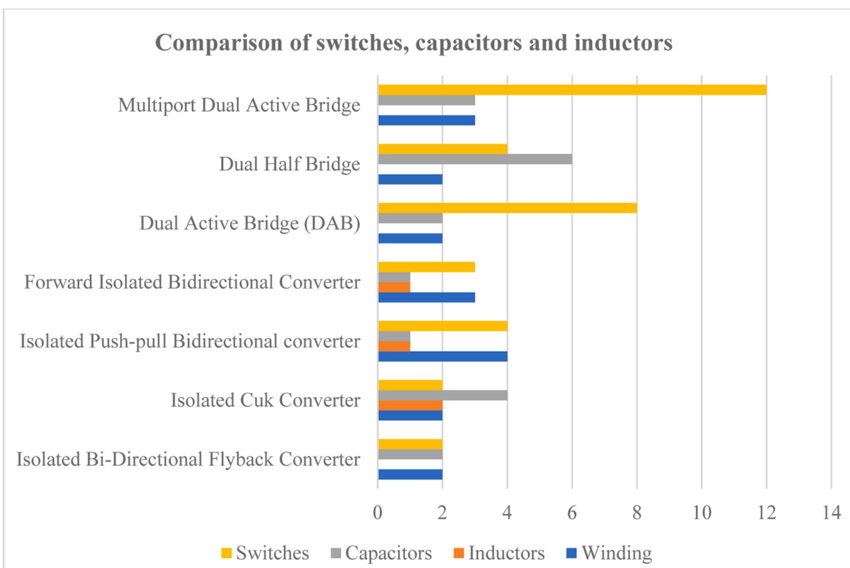

Fig. 7. Comparison of IBDCS in terms of switches, capacitors and inductors.

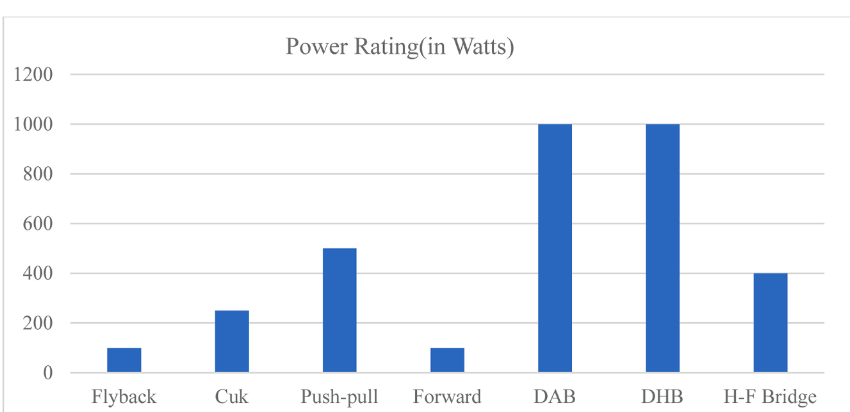

Fig. 8. Power operating range of IBDCs.

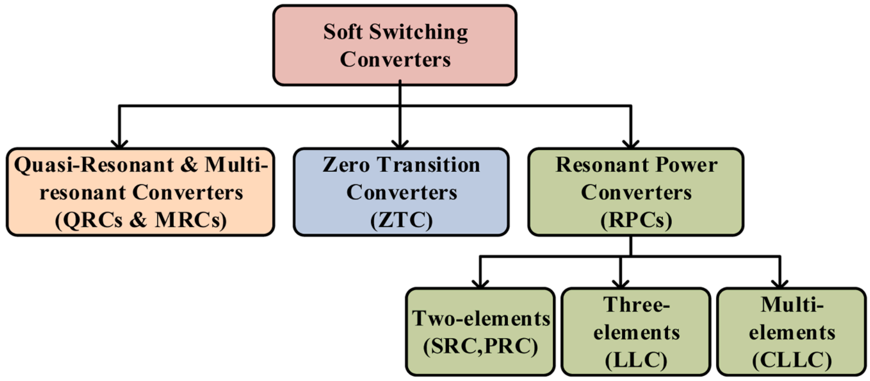

Fig. 9. Classification of Soft Switching/ Resonant Converters.

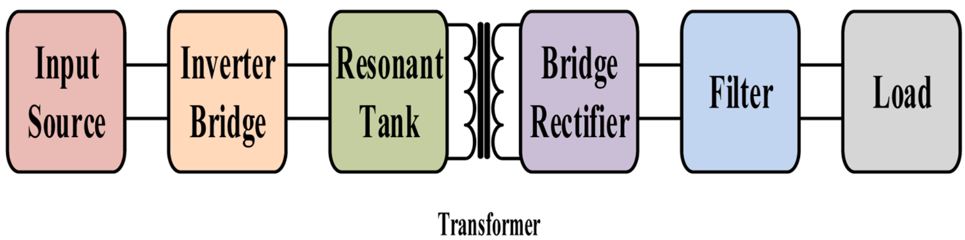

Fig. 10. Working of resonant power converter.

as Qausi-Resonant And Multi-Resonant" Converter (QRC&MRCs), Zero Transition Converters (ZTC) and Resonant Power Converters (RPCs) as shown in Fig. 9. In comparison to linear regulators, these two types possess superior efficiency and the capability to operate at high switching frequencies. This enables the utilization of a compact ferrite transformer core. Additionally, they can function across a broader range of DC input voltages, unlike linear regulators. Bidirectional DC-DC converters have garnered considerable attention in both academic and industrial spheres due to their application in maintaining system reliability and serving as an interface between batteries and super-capacitors, which function as storage devices. The utilization of bidirectional DC-DC power converters is increasingly prevalent across a range of applications that require power flow in both directions. These applications encompass but are not confined to energy storage systems, uninterruptible power supplies, electric vehicles, and renewable energy systems, among others. Currently, resonant DC-DC converters are the favored choice for power conversion in numerous low- and high-voltage applications. Resonant DC-DC converters typically possess attributes that diminish the losses incurred during switching at the inverter switches and output rectifier diodes. This enables them to operate at higher switching frequencies, thereby achieving greater efficiency and ultimately resulting in more compact converters. Due to simplicity and popularity, many researchers have worked on and recommended Resonant Power Converters (RPCs) in various applications. In this paper RPCs are considered for discussion over others. Further RPCs are categorized as two-element (second-order), three-element (third-order) and multi-element (higher order) types of RPCs. Two-element is also termed as second-order resonant tank consists of one inductor and a capacitor. On the basis of source at the input port RPCs can be divided as voltage and current source resonant converters. Working of resonant power converter is shown in Fig. 10 with Input (Voltage/Current) source is connected to the inverter which converter dc into ac by utilizing half/full bridges then output of inverter is provided to Resonant tank. It will act as band pass filter and low/high pass filter also called as Series Resonant Converters (SRCs) and Parallel Resonant Converters (PRCs) respectively; further transformer is added for isolation also provide level up & down of voltage which will be rectified through half/full bridges that passes through Low/High Pass Filter (HPF/LPF) for utilization at load side.

3.1. Two elements resonant converters

3.1.1. Series resonant converters (SRCs)

Series Resonant Converter (SRCs) consists of a bridge inverter, two-element resonant network, an isolation transformer, and a rectifier. The bridge inverter switches are having body diode and parasitic capacitor. The resonance tank circuit contains a resonant inductor (Lr) and a capacitor (Cr) which stores electricity that oscillates with the frequency of the resonant circuit. Flow of energy between the capacitors creates resonance for inductor into LC circuit. The repeated back-and-forth flow of electrical energy between a fully charged inductor and capacitor produces electromagnetic frequency, which is helpful in many applications. It is called SRCs due to reactive elements Lr and Cr are in series

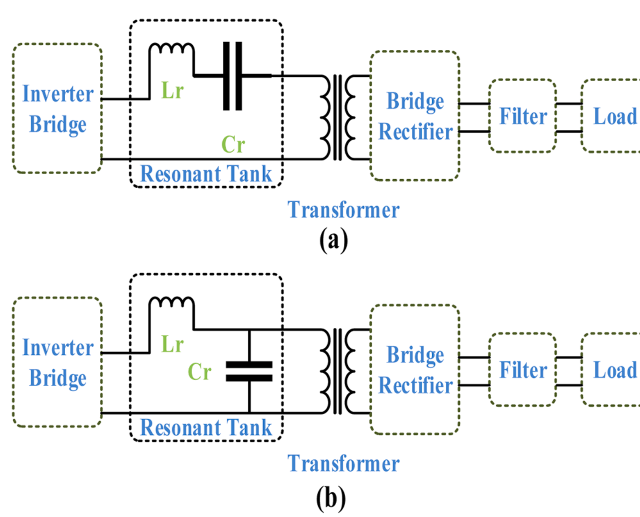

Fig. 11. (a) Series Resonant Converters (SRCs) (b) Parallel Resonant Converters (PRCs).

with transformer's winding as shown in Fig. 11(a). It acts voltage divider due to series connection of resonant tank circuit and load. Also, impedance can be controlled by variating frequency of driving voltage. SRCs are having maximum gain at resonance frequency also capacitor (Cr) is connected in series with transformer which blocks dc components but SRCs cannot operate at no load condition also it produces high ripple current henceforth it cannot be employed for low voltage high current applications [130–132]. Series Resonant Converter (SRCs) is classified under two-element tank converter. A SRC with soft switching operating range is developed with sinusoidal current in both continuous and discontinuous conduction mode but it depends on the passive element [133]. A converter is proposed for two input sources (PV system or battery) along with series resonant converter for LED lighting applications which operates at constant frequency and duty cycle which allows ZVS in wide range, low EMI, etc. [134]. A 3kW DAB dc converter is modelled and designed with series resonant converter by implementing open loop soft starting algorithm which limits high transient through resonant inductor and capacitor [135]. A 40 kW EV charger is proposed with 20 kW modules which is parallelly connected and operates in two stage first with Vienna Rectifier (VR) and second through two Series-Resonant Dual-Active-Bridges (SRDABs) with higher efficiency and improved performance of system [136]. A new sine wave modulation topology has been introduced in [137] where output voltage is adjusted by the duty cycle with fixed frequency and the voltage gain is independent of the load.

3.1.2. Parallel resonant converters (PRCs)

Into Parallel Resonant Converters (PRCs) reactive elements i.e. inductor (Lr) and capacitor (Cr) are connected in parallel to the load; either one element or combination of both will serve same purpose as shown in Fig. 11(b). Due to parallel connection of these reactive elements this configuration is named as Parallel Resonant Converters (PRCs). Output voltage is controlled by variating control switching system frequency. It is classified under two-element tank converter. Also, it overcomes disadvantages of SRCs; At no load condition PRCs are employed to obtain output voltage but PRCs lags where input current increases magnitude of circulating current also increases. New phase-shifted full-bridge topology is introduced with a variable inductor which is connected in parallel near-ideal transformer to provide Full-Load Range ZVS and Low Duty Cycle Loss [138]. Controlling of resonant converter is always a challenge many authors have suggested for Variable frequency modulation but it has some challenges like additional circuitry for EMI filters and switch driver. Authors of [139] has suggested fixed frequency Phase Shift Modulation for controlling of parallel resonant dc-dc converter which results less harmonics.

3.2. Three elements resonant converters

3.2.1. LCC converter

LCC Converter or Series-Parallel Resonant Converters (SPRCs) is combination of series and parallel resonant converters. It consists of three reactive elements i.e., one inductor (Lr) and two capacitor (Cr, Cp) as shown in Fig. 12(a). Series connection of the primary windings reduces the number of turns required in the windings, which helps to reduce the size and weight of the transformers whereas Parallel connection of the secondary windings reduces the current stress on the windings and the conduction losses of the rectifier diodes It overcomes drawback of no-load regulation and higher circulating current in both SRC and PRC respectively. SPRCs regulates output voltage when Cp is not too low value else it will work as Series Resonant Converter [140] propose series-parallel bidirectional isolated dc-dc resonant converters for electric vehicle charging/discharging with frequency modulation technique; to analyze the converter characteristics first-harmonic approximation (FHA) is employed which minimizes circulating current, protection of circuit and delivers efficiency of 97.7 % at the rated power level. Different technique for controlling of Hybrid Electric Vehicle is proposed in [141]. Many authors have proposed wireless charger for EVs but if faces issue of misalignment in coupling coefficient that led to reduction in efficiency [142] has suggested two compensation topologies and implemented on wireless charger prototype of rating 6.1 kW with efficiency of 94 %.

3.2.2. LLC converter

LLC converter is advantageous over others, it is having positive aspects of both SRC and PRC. LLC can be employed in EVs due many reasons like ZVS turn on and off, low harmonics, low voltage stress and wide output range, electrical isolation, high efficiency and energy density [143]. LLC converter is having three reactive elements i.e., two inductor (Lr, Lm) and one capacitor (Cr) which makes this unsymmetrical. Difference between SRC and LLC is presence of magnetizing inductance (Lm), LLC is having three reactive elements i.e., Lr, Lm and Cr where Lr and Cr are resonant inductor and capacitor however Lm is Magnetizing Inductance [144] as shown in Fig. 12(b). A series dual Buck-LLC resonant converter is proposed introducing auxiliary inductors which solves soft switching issue and optimize the energy transmission process also allows for adjusting the duty cycle of series dual buck circuits to be employed for different voltage level [145]. An on-board charger has been proposed and implemented into [146] with

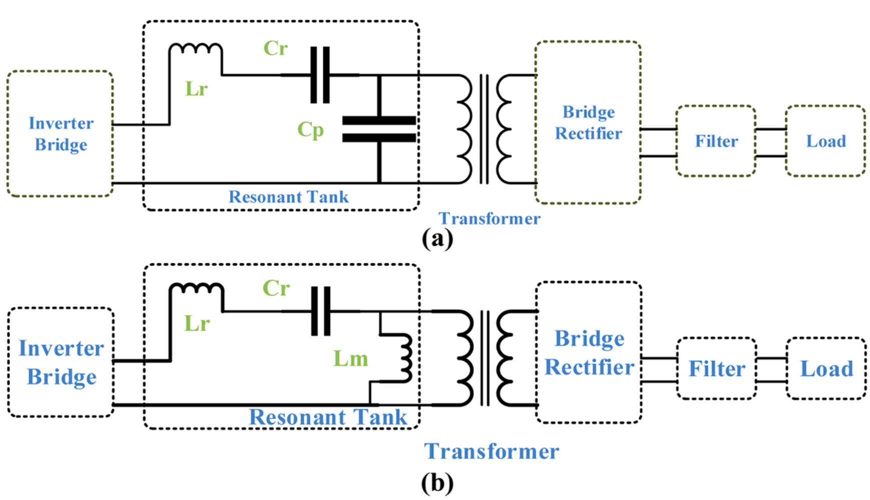

Fig. 12. (a) LCC/Series-Parallel Resonant Converters (SPRCs) (b)LLC Resonant Converters.

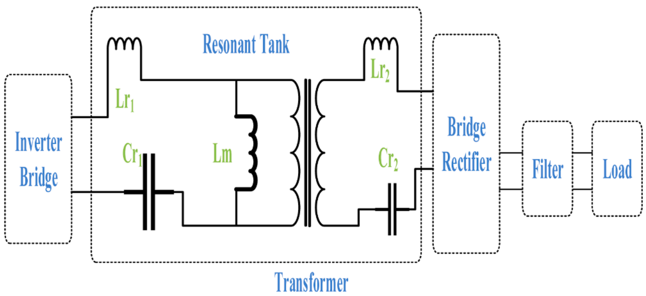

Fig. 13. CLLC Resonant Converters.

full-bridge LLC resonant converter with series-parallel connected transformers. A parallel loaded resonant converter is proposed for wide output voltage application for EVs it offers narrow frequency range soft switching irrespective of voltage and load variation [147]. In [148] a high voltage gain converter is proposed with low duty cycle ration. A novel dual-bridge LLC resonant converter with wide range of low input voltage is proposed by the controlling of phase shift modulation topology also with different duty ratios which led to higher efficiency [149].

3.3. Multi elements resonant converters

3.3.1. CLLC converter

Into CLLC resonant converter four reactive elements (Lr1, Cr1 Lr2, Cr2) are present. Also, there is no voltage drop on the primary side Lr1,

Table 12
Merit, Demerit and major application of NBDCs, IBDCs and Soft Switching Converter.

Reference	Configuration	Topology	Merit	Demerit	Major Applications
[5,6]	Non-Isolated Bidirectional Converter (NBDCs)	Buck-boost	Wide range of input voltage, Lower Heat Dissipation	Additional filter for Noise & Ripple	Renewable Energy Systems (RESs), Automobile Electronics, Battery-Powered Devices
[7,14]	Non-Isolated Bidirectional Converter	Cuk	Continuous Input and Output Current, Balanced Power	Higher Stress level, design issue at higher frequency	Power Factor Corrector (PFC), DC-Microgrid, EV charger
[8]		SEPIC/Zeta	Reduced ripple currents, increased voltage gain and continuous input current with switching stress	Coupled inductors reduces efficiency	Energy Storage System (ESS), Distributed Power System
[9,10,17]		Switched Capacitor	Reduces Electro Magnetic Interference (EMI)	Continuous Input Current, losses at junction capacitors and forward voltages	Rechargeable batteries, Rail supplies
[10,11]		Cascaded	Reduces output current ripple, improved efficiency of 98 % with low output voltage ripple	Inductor loss effects boost ratio, challenges while balancing of voltage	Motor control of EVs, Solar Photovoltaic system (PV)
[19]		Interleaved	Soft-charging, auto-current sharing mechanisms	Surge currents, Significant switching and conduction losses	Energy Storage System (ESS), Battery Chargers
[13,20]		Multilevel	Self-balanced voltage	boost ratio limitation due to inductor losses	MLI for Renewable Energy Systems (RESs)
[29]	Isolated Bidirectional Converter (IBDCs)	Flyback	Improved efficiency, improved energy recovery of leakage inductance	Limited output voltage due to parasitic capacitance, Discontinuous input current	Solar Photovoltaic system (PV), Thermoelectric energy generation system
		Cuk	Reduced components and lower design cost	Resonance causes more stress on switches, Limited voltage gain	DC microgrid applications, EV Charger, Power tools
[52,54, 55]		Push-pull	Reduced weight and size, Minimized switching loss, High efficiency & conversion ratio	Issue with High Voltage range and less power transfer capability at specific duty cycle	Energy Storage System (ESS), DC-Microgrid
[65]		Forward	Minimized switching loss, low voltage stress	High circuit complexity, bulky filters	Solar applications, Power supply for digital devices, Drivers circuit
[88]		Dual Active Bridge (DAB)	Improved dynamic response in various conditions, high conversion efficiency and good performance in steady state	Soft-switching failure into closed-loop systems, control performance is affected during transients, Issue of dc-bias	Solar Photovoltaic system (PV), Utility Interfacing, Electric Vehicle (EVs), DC microgrids, Energy Storage Systems (ESS)
[94]		Dual Half Bridge (DHB)	Soft Switching at higher voltage gain and low voltage stress, less nos. of power switches,	Traditional control methods decrease power factor. High Input current ripple	Microgrids, UPS
[106,107] [111, 112]		Half-Full Bridge (HFB)	Reverse power flow capability, minimized switching loss	Higher operating frequencies causes stress in switches, Limited ZVS range	Battery & EVs charger, Residential photovoltaic system
[119] [123, 125]		Multiport Dual Active Bridge (DAB)	Improved efficiency and voltage conversion ratio	Unequal load sharing in converters with multiple sources, Difficulty in High Frequency Transformer (HFT) insulation design	Future aircraft, PV based aerospace, fuel cells and hybrid energy storage.
[136]	Soft Switching Converters	Series Resonant Converters (SRCs)	Maximum gain at resonance frequency, Block dc component, low EMI	Can't operate at no load condition, high ripple current	EV charger, LED lighting, Induction heating, RF power supplies, medical equipment
		Parallel Resonant Converters (PRCs)	Full load ZVS range, low duty cycle loss	Complex controlling methods, additional filters and driver circuit, high switch off current and high circulating energy	High-voltage power supplies, hybrid electric vehicles (HEVs)
[140,141]		LCC Converter	Reduced stress, Improved efficiency for wide load range, soft switching	Complex controlling methods, misalignment in coupling coefficient	Server power supplies, LED lighting, hybrid electric vehicles (HEVs)
[146] [147]		LLC Converter	low harmonics, low voltage stress and wide output range, electrical isolation, high efficiency	additional components and circuit complexity	Electric vehicles charger, Renewable Energy Systems, Telecommunications
[150] [151]		CLLC Converter	Less reactive current. wide range of ZVS, switching losses	Increased complexity, cost and size	Plug in Electric Vehicles (PHEVs), Power Factor Correction (PFC)

Cr1 and secondary side Lr2, Cr2 as shown in Fig. 13. Thus, the fundamental harmonics of the primary and secondary sides are equal which indicates constant voltage gain ability of CLLC resonant converter. It is having enhanced soft switching area also it offers higher efficiency. In [150] a CLLC resonant converter with 97.85 % efficiency is proposed. It is implemented for both half, full bridge for plug in electric vehicles and a comparison has been done in [151]. [152] discussed optimum design of CLLC resonant converter and also performed mode analysis but issue with design is lack of synchronous rectification which led to increased conduction loss on secondary side; this issue is resolved into [153–157] with the implementation of two different control loop.

4. Conclusion

This paper investigates and reviews the bidirectional dc converter from its classification to the topological structure. It can also be analyzed from distinct viewpoints; this research primarily provides an analysis of these converters from the perspective of topology and bridge structure also. Table 1 and Fig. 4 is included which shows a comparison of NBDCs with switches, inductors, capacitors, windings count, power rating and application. As discussed, buck and boost derived converter are used where different voltage level has to be achieved as compared to an input voltage. Cuk converter provides continuous input and output current but it faces the issue of current ripples which is eliminated by using a coupled inductor. When this cuk converter is rearranged, new converter is obtained i.e. SEPIC and ZETA it has a high voltage (V_H) and low voltage (V_L) in both paths but with the same polarities; these are the simple bridge arrangement voltage that can be boosted by using one more technique called voltage boosting techniques. These techniques are used in the switched-capacitor converter which uses a capacitor instead of using an inductor. By cascading, the voltage can be boosted in a manner by providing an output of one to the input of others interleaved and multilevel converters are also the solution to the increased output voltage. IBDC is also classified into these two topologies i.e., basic and bridge. For low-power application, a flyback converter is preferred but it faces the issue of ringing, leakage inductance and ripples which is eliminated by using a snubber and ripple cancelling circuit respectively. Cuk converter is made up of a capacitor and inductor for higher voltage gain which makes it suitable for higher power applications but it faces the issue of leakage inductance. To overcome these issues many methods are used like soft switching, snubber circuits etc. For the range of 200W-500 W forward converters are used but it faces the issue of leakage in magnetizing current. To overcome these issues many hybrid configurations of the converter are discussed. When two forward converters are connected back-to-back, this arrangement is called a push-pull converter it utilizes magnetization of the core in both directions many configurations are discussed which utilize transformer action and also double voltage. Dual Active Bridge (DAB) is an arrangement of two bridges with inversion and rectification modes integrated with a high-frequency transformer and operated by providing phase shift between these bridges. For the application up to 500 W, Dual Half Bridge (DHB) is preferred it consists of two half-bridges with high-frequency transformer whose performance is enhanced and its ZVS range gets wider which also reduces conduction losses. A converter is proposed for providing the current balance between batteries with a half-full bridge linked to a high-frequency transformer i.e., Half-Full Bridge (HFB). This HFB converter is employed mainly in HVDC, Battery balancers, Modular Multilevel converters etc. Integrating multiple input sources needs a multiport converter, a Multiport Dual Active Bridge (DAB) is developed for the integration of these multiple input voltage sources without using additional circuits. These bidirectional converters are employed from a few watts to hundred kilowatts; Table 10 and Fig. 7 is included to summarized the comparison of IBDCS in terms of switches, capacitors, inductors Also, Table 11 shows advantages and disadvantages of IBDCs. In last section soft switching converters has been discussed mainly tending towards implementation into EVs. Different converter is discussed for the selection of Resonant Power Converters for EVs and chargers at different rating. CLLC is found more suitable for the Electric Vehicle application. Accompanying difficulties with EV charges based on resonant converters are emphasized, and their potential future growth is also discussed. Fig. 8 shows different power operating range of IBDCs and Table 12 is summary of all converter in terms of merits, demerits and applications; which will help author to select suitable converter for their applications.

CRediT authorship contribution statement

Mohammad Aslam Alam: Writing – original draft, Conceptualization, Formal analysis, Investigation, Data curation, Writing – review & editing. **Ahmad Faiz Minai:** Conceptualization, Methodology, Formal analysis, Investigation, Supervision. **Farhad Ilahi Bakhsh:** Conceptualization, Methodology, Formal analysis, Investigation, Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Mohammad Aslam Alam has completed Bachelor of Technology in Electrical & Electronics Engineering, Integral University, Lucknow (U.P.) India and MTech in Power Electronics & Drives from Madan Mohan Malaviya University of Technology, Gorakhpur (U.P.) India. Currently he is pursuing PhD in Electrical Engineering from Integral University, Lucknow (U. P.) India. His-area of research is Bidirectional DC-DC converter for hybrid system.

Dr. Ahmad Faiz Minai received bachelor's degree in Electrical Engineering from Uttar Pradesh Technical University (Currently it is AKTU) in 2007, Master's degree in Electrical Engineering from ZHCET, Aligarh Muslim University in 2009 and Doctorate from Integral University, Lucknow in 2017. He is currently working as an Associate Professor in the Department of Electrical Engineering, Faculty of Engineering, Integral University, Lucknow, India. His-research areas include Power Electronics Converters, Soft Computing in Renewable Power Generation, Solar Photovoltaics, Energy Demand Modelling and Internet of Things. He has many research publications in International Conferences and reputed refereed SCIE/ESCI/SCOPUS Journals.

Dr Farhad Ilahi Bakhsh received Diploma and B. Tech degree in Electrical Engineering from Aligarh Muslim University (AMU), Aligarh, India in 2006 and 2010, respectively. He was awarded University Medal (Gold) for standing first throughout Diploma in Electrical Engineering. Then he pursued Masters in Power System and Drives from the same University. In Masters he secured first position in his branch. He joined IEEE during Masters and since then he is an IEEE member. He also worked as head of Research & Development cell, IEEE student chapter, AMU for around two years. Under this cell, he developed five new systems i.e. A rotor power control based flexible asynchronous AC link (FASAL) system, A miss-call based switching system for multiple loads or appliances, A power controller circuit based flexible asynchronous AC link (FASAL) system for induction generator applications, A combined voltage control and rotor power control based flexible asynchronous AC link (FASAL) system and A waste fluid pressure-based energy generation system. Among these five systems, four system has been published by an official Journal of Patent Office.

Then he pursued Ph.D. from Indian Institute of Technology Roorkee, India. During his Ph.D. he developed a new method for grid integration for wind energy generation system which has been recognized worldwide. He served as Assistant Professor in Department of Electrical & Renewable Energy Engineering, School of Engineering & Technology, Baba Ghulam Shah Badshah University, Rajouri, J & K, India. He developed an automatic solar tracking system which has been appreciated by IEEE India Council, Centre for Embedded Product Design, Centre for Electronics Design and Technology, Netaji Subhas Institute of Technology in association with IEEE Delhi Section & IEEE CAS, Bangalore Chapter. Currently he is serving as Assistant Professor in Department of Electrical Engineering, National Institute of Technology Srinagar, Jammu & Kashmir, India. He is founder and Counselor of IEEE Student Branch, NIT Srinagar.